

PUBLICATION LIST - FULL

Ulyana Dupletsa

1. “The geometry of lunar gravitational wave detection”
J. Tissino, F. Santoliquido, F. Iacovelli, **U. Dupletsa**, T. D. Canton, M. Ballelli, A. Chopra, L. E. Espinosa Castro, L. Pezzella, M. Schulz, I. Takimoto Schmiegelow, J. Harms; *submitted to the Lunar Gravitational Wave Detection Nature Collection*, arXiv:2606.04918 [astro-ph.IM] [gr-qc]
2. “Radio sirens: inferring H_0 with binary black holes and neutral hydrogen in the era of the Einstein Telescope and the SKA Observatory”
U. Dupletsa, S. Mastrogiovanni, M. Spinelli, T. Ronconi, M. Schulz, R. Murgia, J. Harms, T. Baker, M. Calabrese, C. Carbone, S. Cunningham, I. Harrison, K. Leyde, D. Nanadoumgar-Lacroze; *submitted to MNRAS*, arXiv:2605.12606 [astro-ph.CO] [astro-ph.HE]; Zenodo dataset, <https://doi.org/10.5281/zenodo.20582389>; GitHub 
3. “Probing Cosmic Expansion and Early Universe with Einstein Telescope”
A. Ricciardone et al. incl. **U. Dupletsa**; *ESO Expanding Horizon White Paper*, arXiv:2601.06017 [astro-ph.CO] [gr-qc]
4. “Comparing next-generation detector configurations for high-redshift gravitational wave sources with neural posterior estimation”
F. Santoliquido, J. Tissino, **U. Dupletsa**, M. Branchesi, J. Harms; *Astronomy & Astrophysics*, 708 (2026) A175, arXiv:2512.20699 [gr-qc] [astro-ph.HE]
5. “Towards an anomaly detection pipeline for gravitational waves at the Einstein Telescope”
G. Inguglia, H. Haigh, K. Vitulova, **U. Dupletsa**; *Physics Letters B*, 874 (2026), arXiv:2511.13154 [gr-qc] [astro-ph.IM]
6. “Multi-messenger observations of binary neutron star mergers: synergies between the next generation gravitational wave interferometers and wide-field, high-multiplex spectroscopic facilities”
S. Bisero, D. Vergani, E. Loffredo, M. Branchesi, H. Nandini, **U. Dupletsa**, R. I. Anderson; *Astronomy & Astrophysics*, 705 (2026) A54, arXiv:2507.02055 [astro-ph.HE]
7. “Fast and accurate parameter estimation of high-redshift sources with the Einstein Telescope”
F. Santoliquido, J. Tissino, **U. Dupletsa**, M. Branchesi, J. Harms, M. Arca Sedda, M. Dax, A. Kofler, S. R. Green, N. Gupte, I. M. Romero-Shaw, E. Berti; *PRD*, 112 (2025) 10, arXiv:2504.21087 [astro-ph.HE] [astro-ph.IM] [gr-qc]
8. “The Science of the Einstein Telescope”
A. Abac et al. incl. **U. Dupletsa**; *JCAP* 03 (2026) 081, arXiv:2503.12263 [gr-qc]
9. “Blinded Mock Data Challenge for Gravitational-Wave Cosmology-I: Assessing the Robustness of Methods Using Binary Black Holes Mass Spectrum”
A. Agarwal, **U. Dupletsa**, K. Leyde, S. Mukherjee, B. Revenu, J. E. Rivera, A. E. Romano, M. R. Sah, S. Vallejo-Pena, A. Avendano, F. Beirnaert, G. Dalya, M. C. Espitia, C. Karathanasis, S. Moreno-Gonzalez, L. Quiceno, F. Stachurski, J. Garcia-Bellido, R. Gray, N. Tamanini, C. Turski; *The Astrophysical Journal*, Vol. 987 (2025), DOI:10.3847/1538-4357/adda3a, arXiv:2412.14244 [astro-ph.CO], GitHub 
10. “Model-independent cosmology with joint observations of gravitational waves and γ -ray bursts”
A. Cozzumbo, **U. Dupletsa**, R. Cald eron, R. Murgia, G. Oganesyan, M. Branchesi; *JCAP*, Vol. 05 (2025) 021, DOI:10.1088/1475-7516/2025/05/021, arXiv:2411.02490 [gr-qc]
11. “Prospects for optical detections from binary neutron star mergers with the next-generation multi-messenger observatories”
E. Loffredo, N. Hazra, **U. Dupletsa**, M. Branchesi, S. Ronchini, F. Santoliquido, A. Perego, B. Banerjee, S. Bisero, G. Ricigliano, S. Vergani, I. Andreoni, M. Cantiello, J. Harms, M. Mapelli, G. Oganesyan; *Astronomy & Astrophysics*, Vol. 697 (2025) A36, DOI:10.1051/0004-6361/202452863, arXiv:2411.02342 [astro-ph.HE]; Zenodo dataset <https://zenodo.org/records/13850416>
12. “Validating Prior-informed Fisher-matrix Analyses against GWTC Data”
U. Dupletsa, J. Harms, Ken K. Y. Ng, J. Tissino, F. Santoliquido, A. Cozzumbo; *Physical Review D*, Vol. 111 (2025), 2, 024036, DOI:10.1103/PhysRevD.111.024036, arXiv:2404.16103 [gr-qc], GitHub 
13. “Classifying binary black holes from Population III stars with the Einstein Telescope: a machine-learning approach”
F. Santoliquido, **U. Dupletsa**, J. Tissino, M. Branchesi, F. Iacovelli, G. Iorio, M. Mapelli, D. Gerosa, J. Harms and M. Pasquato; *Astronomy & Astrophysics*, Vol. 690 (2024), A362, DOI:10.1051/0004-6361/202450381, arXiv:2404.10048 [astro-ph.HE]
14. “The Wide-field Spectroscopic Telescope (WST) Science White Paper”
V. Mainieri et al. incl. **U. Dupletsa**; 2024, arXiv:2403.05398 [astro-ph.IM]
15. “Phenomenological models of Cosmic Ray transport in Galaxies”
C. Evoli and **U. Dupletsa**; 2023, in *Proceedings of the International School of Physics “Enrico Fermi”*, Vol. 208: Foundations of Cosmic Ray Astrophysics, arXiv:2309.00298 [astro-ph.HE]

16. “*Science with the Einstein Telescope: a comparison of different designs*”
M. Branchesi, M. Maggiore et al. incl. **U. Dupletsa**; *JCAP*, Vol. 07 (2023) 068, DOI:10.1088/1475-7516/2023/07/068, arXiv:2303.15923 [gr-qc] [astro-ph.CO] [astro-ph.HE]
17. “*Pre-merger alert to detect the very-high-energy prompt emission from binary neutron-star mergers: Einstein Telescope and Cherenkov Telescope Array synergy*”,
B. Banerjee, G. Oganesyan, M. Branchesi, **U. Dupletsa**, F. Aharonian, F. Brighenti, B. Goncharov, J. Harms, M. Mapelli, S. Ronchini, F. Santoliquido
Astronomy & Astrophysics, Vol. 678, DOI:10.1051/0004-6361/202345850,
arXiv:2212.14007 [astro-ph.HE]
18. “*Measuring properties of primordial black hole mergers at cosmological distances: effect of higher order modes in gravitational waves*”,
K. K. Y. Ng, B. Goncharov, S. Chen, S. Borhanian, **U. Dupletsa**, G. Franciolini, M. Branchesi, J. Harms, M. Maggiore, A. Riotto, B. S. Sathyaprakash, S. Vitale, *Physical Review D*, Vol. 107, 024041,
DOI:10.1103/PhysRevD.107.024041, arXiv:2210.03132 [astro-ph.HE] [gr-qc]
19. “*GWFish: A simulation software to evaluate parameter-estimation capabilities of gravitational-wave detector networks*”
U. Dupletsa, J. Harms, B. Banerjee, M. Branchesi, B. Goncharov, A. Maselli, A. C. S. Oliveira, S. Ronchini, J. Tissino; *Astronomy and Computing*, Vol. 42 (2023), DOI:10.1016/j.ascom.2022.100671, arXiv:2205.02499 [gr-qc], GitHub 
20. “*Perspectives for multi-messenger astronomy with the next generation of gravitational-wave detectors and high-energy satellites*”,
S. Ronchini, M. Branchesi, G. Oganesyan, B. Banerjee, **U. Dupletsa**, G. Ghirlanda, J. Harms, M. Mapelli, F. Santoliquido; *Astronomy & Astrophysics*, Vol. 665 (2022), A97, DOI:10.1051/0004-6361/202243705,
arXiv:2108.07276 [astro-ph.HE]
21. “*On the single-event-based identification of primordial black hole mergers at cosmological distances*”,
K. K. Y. Ng, S. Chen, B. Goncharov, **U. Dupletsa**, S. Borhanian, M. Branchesi, J. Harms, M. Maggiore, B. S. Sathyaprakash, S. Vitale; *ApJL*, Vol. 931 (2022) L12, DOI 10.3847/2041-8213/ac6bea, arXiv:2108.07276 [gr-qc], [hep-ph]

COLLABORATION PAPERS

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22. “*GWTC-4.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation*”
LIGO-Virgo-KAGRA Collaboration; arXiv:2509.04348 [astro-ph.CO], accepted in *APJ Letters*
[Member of analysis and paper writing teams]
 23. “*GWTC-5.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation*”
LIGO-Virgo-KAGRA Collaboration; arXiv:2605.27227 [astro-ph.CO]
[Member of analysis team]
 24. **50+ Collaboration Papers inside the LIGO-Virgo-KAGRA Collaboration** [see complete list [here](#)]